

On the isolation number of graphs with minimum degree four

^{1,2}Wayne Goddard and ²Michael A. Henning

¹School of Mathematical and Statistical Sciences
Clemson University, United States of America
Email: `goddard@clemson.edu`

²Department of Mathematics and Applied Mathematics
University of Johannesburg, South Africa
Email: `mahenning@uj.ac.za`

Abstract

An isolating set in a graph G is a set S of vertices such that removing S and its neighborhood leaves no edge. The isolation number of G (also known as the vertex-edge domination number), denoted $\iota(G)$, is the minimum size among all isolating sets of G . We provide a technique for proving upper bounds on this parameter for graphs with a given minimum degree. For example, we show that if G has order n and minimum degree at least 4, then $\iota(G) \leq 13n/41$, and if G is also triangle-free, then $\iota(G) \leq 3n/10$.

Keywords: Isolation number; Minimum degree

AMS subject classification: 05C69

1 Introduction

As defined by Caro and Hansberg [7], an isolating set in a graph G is a set S of vertices such that removing S and its neighborhood leaves no edge. The isolation number of G , denoted $\iota(G)$, is the minimum size among all isolating sets of G . This parameter is equivalent to the vertex-edge domination number, introduced earlier by Peters [12]. A fundamental upper bound on the parameter was provided by Caro and Hansberg and by Żyliński:

Proposition 1 [7, 14] *If G is a connected graph of order n other than K_2 and C_5 , then $\iota(G) \leq n/3$.*

The extremal trees were characterized by Krishnakumari et al. [11] and Dapena et al. [8], and the extremal graphs were characterized in [1]. Apart from one graph, all the extremal graphs have a vertex of degree two or less, and there are infinitely many extremal graphs with minimum degree 2.

A natural question then is what happens for larger minimum degree. In general, Caro and Hansberg [7] showed that the isolation number of a graph of order n is at most $(\ln(\delta + 1) + \frac{1}{2})n/(\delta + 1)$ if every vertex has degree at least δ . Also, they showed there are arbitrarily large connected graphs with minimum degree δ and isolation number $2n/(\delta + 4)$. Another result in this direction was provided by Ziemann and Żyliński [13], who showed that apart from one graph of order 6, the isolation number of a cubic graph is at most $10n/31$. They also showed that there are arbitrarily large cubic graphs with isolation number $2n/7$.

In this paper we introduce a tool called the isolation residual graph, modeled after the residual graph used in bounds on the game domination number (see for example [2, 10]) and the domination number (see for example [3, 4, 5, 6]). Using this tool we provide several upper bounds for the isolation number of a graph given a lower bound on its minimum degree. For example, we show that if G is a graph of order n with minimum degree at least 4, then $\iota(G) \leq 13n/41$, and if G is also triangle-free, then $\iota(G) \leq 3n/10$.

We proceed by first introducing the isolation residual graph and then proving the upper bounds. Thereafter we briefly discuss associated lower bounds for these problems.

2 Methodology

We establish upper bounds on the isolation number of a graph with given minimum degree. For notation and graph theory terminology, we generally follow [9]. Specifically, let G be a graph with vertex set $V(G)$ and edge set $E(G)$. For a set of vertices $S \subseteq V(G)$, the subgraph induced by S is denoted by $G[S]$. For set D of vertices let $N[D]$ be the closed neighborhood of D , that is, the union of D and its neighbors.

Our proof is based on the proof technique employed to establish upper bounds on the game domination number or domination number of a graph. That proof uses a *residual graph*, which is defined by coloring and weighting the vertices of the graph as to how they relate to a given partial dominating set. We extend this:

Definition 1 Given a graph G and a subset $D \subseteq V(G)$, the *isolation residual graph* denoted by G_D^ι is obtained as follows.

- (a) A vertex in $G - N[D]$ with a neighbor in $G - N[D]$ is colored *white*.
- (b) A vertex in $N[D]$ with a white neighbor is colored *blue*.

- (c) All other vertices are colored *red*.
- (d) The edge set of G_D^ℓ consists of all edges of G that are incident with at least one white vertex.

In other words, the white vertices still need to be isolated, the blue vertices might be useful to do this isolating, and the red vertices are no longer relevant. We denote the set of red vertices by R , the set of blue vertices by B , and the set of white vertices by W . We observe that these sets form a (weak) partition of $V(G)$. Note that:

Observation 1 *The following holds in the isolation residual graph G_D^ℓ :*

- (a) *Every vertex of D is colored red.*
- (b) *Every red vertex is isolated, but no white nor blue vertex is isolated.*
- (c) *The degree of a white vertex in G_D^ℓ equals its degree in G .*
- (d) *The set D is an isolating set in G if and only if there is no white vertex.*

We now introduce a weight function (or potential function). We define B_4 as the set of blue vertices of degree at least 4 in G_D^ℓ , and for $1 \leq i \leq 3$ we define B_i as the set of blue vertices of degree exactly i in G_D^ℓ . The weight $w(v)$ of a vertex v is defined as:

$$w(v) = \begin{cases} \omega & \text{if } v \in W, \\ \beta_i & \text{if } v \in B_i \text{ for } i \in \{1, 2, 3, 4\}, \text{ and} \\ 0 & \text{if } v \in R, \end{cases}$$

where ω and each β_i are positive constants to be determined later. We define the weight of the isolation residual graph G_D^ℓ as

$$w(G_D^\ell) = \sum_{v \in V} w(v).$$

By Observation 1, $w(G_D^\ell) = 0$ if and only if D is an isolating set in G .

The proof strategy is to start with $D = \emptyset$ and repeatedly enlarge the set D . We need the following immediate observation.

Observation 2 *Suppose $D \subseteq D' \subseteq V(G)$. Every red vertex in G_D^ℓ is red in $G_{D'}^\ell$. Every blue vertex in G_D^ℓ is red or blue in $G_{D'}^\ell$.*

For set $A \subseteq V(G) \setminus D$ and $D' = D \cup A$, define

$$\xi(A) = w(G_D^\ell) - w(G_{D'}^\ell)$$

as the decrease of the weight when extending D to D' . We say that such a set A is a *D-desirable set* if

$$\xi(A) \geq |A|.$$

We say a weighted graph G is w -alluring if whenever $D \subseteq V(G)$ is not an isolating set of G , the graph G contains a D -desirable set A .

Theorem 1 *If there is a setting w of the weights such that graph G is w -alluring, then $\iota(G) \leq \omega n$ where n is the order of G .*

Proof. Let $D_0 = \emptyset$. In the isolation residual graph $G_{D_0}^\iota$, every vertex is white and hence $w(G_{D_0}^\iota) = \omega n$. Since G is w -alluring, there exists a D_0 -desirable set A_1 . Let $D_1 = A_1$. If D_1 is not an isolating set of G , then there exists a D_1 -desirable set A_2 . Let $D_2 = A_1 \cup A_2$. If D_2 is not an isolating set of G , then we continue the process. At the end we obtain an isolating set $D = D_k = A_1 \cup \dots \cup A_k$ of G such that

$$0 = w(G_{D_k}^\iota) \leq w(G_{D_{k-1}}^\iota) - |A_k| \leq \dots \leq w(G_{D_0}^\iota) - \sum_{i=1}^k |A_i| = \omega n - |D|.$$

Thus $\iota(G) \leq |D| \leq \omega n$. $\square$

Thus to provide an upper bound for a family of graphs, it suffices to find a setting w of the weights such that every graph in that family is w -alluring.

3 System of Constraints

In this section we construct a set of constraints such that any solution w is w -alluring. It is natural that the weight of a blue vertex should not increase as the number of its white neighbors decreases. Thus we specify that $\omega \geq \beta_4 \geq \beta_3 \geq \beta_2 \geq \beta_1 > 0$. Furthermore, for convenience we define

$$\varepsilon_1 = \beta_1 \quad \text{and} \quad \varepsilon_i = \beta_i - \beta_{i-1} \quad \text{for } 2 \leq i \leq 4,$$

and restrict to the case that ε_i does not decrease as i decreases.

Consider the following list $\mathcal{L}$ of constraints:

$$\begin{aligned}
\omega &\geq \beta_4 \geq \beta_3 \geq \beta_2 \geq \beta_1 > 0 && \text{(assumed)} \\
\varepsilon_4 &\leq \varepsilon_3 \leq \varepsilon_2 \leq \varepsilon_1 = \beta_1 && \text{(assumed)} \\
\omega + 5(\omega - \beta_4) &\geq 1 && \text{(Claim 1)} \\
\omega + 4(\omega - \beta_3) &\geq 1 \\
\beta_4 + 5(\omega - \beta_3) &\geq 1 \\
\omega + 3(\omega - \beta_2) + 4(\delta - 3)\varepsilon_4 &\geq 1 && \text{(Claim 2)} \\
\beta_4 + 4(\omega - \beta_2) + 4(\delta - 3)\varepsilon_4 &\geq 1 \\
\omega + (\delta - 2)\varepsilon_3 &\geq \frac{1}{3} && \text{(Claim 3)} \\
4\omega + \beta_2 &\geq 1 && \text{(Claim 4)} \\
7\omega + \beta_2 &\geq 2 \\
10\omega + \beta_2 &\geq 3 \\
2\omega + 2(\delta - 1) \min(\beta_1, \beta_2/2) &\geq 1 && \text{(Claim 5)} \\
5\omega + 5(\delta - 2) \min(\beta_1, \beta_2/2, \beta_3/3) &\geq 2
\end{aligned}$$

We prove:

Lemma 1 *Assume the weights w satisfy the system of constraints $\mathcal{L}$ for some $\delta \geq 3$. Then every graph with minimum degree at least δ is w -alluring.*

We prove Lemma 1 by a series of claims, where each claim shows that, if some condition holds, a D -desirable set is guaranteed to exist.

Let Δ_W denote the maximum number of white neighbors of a white vertex in G_D^t , and Δ_B denote the maximum degree of a blue vertex.

Claim 1 *We may assume $\Delta_W \leq 3$ and $\Delta_B \leq 4$.*

Proof. (i) Assume $\Delta_W \geq 4$. Let $A = \{v\}$ where v is a white vertex with Δ_W white neighbors. In G_D^t , the vertex v is recolored red. Every white neighbor u of v is recolored blue or red. Since u has at most $\Delta_W - 1$ white neighbors in G_D^t , the weight decrease of u is at least $\omega - \beta_{\Delta_W - 1}$. Hence $\xi(A) \geq \omega + 5(\omega - \beta_4)$ if $\Delta_W \geq 5$, and $\xi(A) \geq \omega + 4(\omega - \beta_3)$ if $\Delta_W = 4$. Thus the set A is a D -desirable set provided

$$\begin{aligned}
\omega + 5(\omega - \beta_4) &\geq 1, \text{ and} \\
\omega + 4(\omega - \beta_3) &\geq 1.
\end{aligned}$$

(ii) Assume $\Delta_B \geq 5$. Let $A = \{w\}$ where w a blue vertex with Δ_B (white) neighbors. Since w becomes red, the weight decrease of w is β_4 . Let x be an arbitrary (white) neighbor of w in G_D^t .

Since by part (i) we may assume $\Delta_W \leq 3$, the weight decrease of x is at least $\omega - \beta_3$. Hence $\xi(A) \geq \beta_4 + 5(\omega - \beta_3)$. Thus the set A is a D -desirable set provided

$$\beta_4 + 5(\omega - \beta_3) \geq 1. \quad (\square)$$

It follows that we may assume that a vertex in B_4 has degree exactly 4. Hence: if $y \in B_i$ has an incident edge e in G_D^ℓ (necessarily joining it to a white vertex) and edge e is removed, then y has exactly $i - 1$ white neighbors. Thus the decrease in weight of y is at least ε_i .

Claim 2 *We may assume $\Delta_W \leq 2$ and $\Delta_B \leq 3$.*

Proof. (i) By Claim 1 we may assume $\Delta_W \leq 3$. Assume $\Delta_W = 3$. Let $A = \{v\}$ where v is a white vertex of with Δ_W white neighbors. In G_D^ℓ , the vertex v is recolored red. Every white neighbor u of v is recolored blue or red, and has at most $\Delta_W - 1 = 2$ white neighbors in G_D^ℓ . Thus the weight decrease of u is at least $\omega - \beta_2$. By the minimum degree, every white vertex has at least $\delta - 3$ edges to B . Thus there are at least $4(\delta - 3)$ edges joining B to v and its white neighbors. The recoloring causes each of these edges to be removed and thus decreases the total weight of B by at least $\min(\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4) = \varepsilon_4$ for each such edge. Hence $\xi(A) \geq \omega + 3(\omega - \beta_2) + 4(\delta - 3)\varepsilon_4$ in this case. Thus the set A is a D -desirable set provided

$$\omega + 3(\omega - \beta_2) + 4(\delta - 3)\varepsilon_4 \geq 1.$$

(ii) By Claim 1 we may assume $\Delta_B \leq 4$. Assume $\Delta_B = 4$. Let $A = \{w\}$ where w is a blue vertex of degree Δ_B . We note that the weight decrease of w is β_4 . Let x be an arbitrary (white) neighbor of w in G_D^ℓ . Since by part (i) we may assume $\Delta_W \leq 2$, the weight decrease of x is at least $\omega - \beta_2$. Further, vertex x has at least $\delta - 3$ blue neighbors different from w . Thus, there are at least $\Delta_B(\delta - 3)$ edges joining $B \setminus \{w\}$ to the white neighbors of w . The removal of each such edge decreases the total weight of $B \setminus \{w\}$ by at least $\min(\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4) = \varepsilon_4$. Hence $\xi(A) \geq \beta_4 + 4(\omega - \beta_2) + 4(\delta - 3)\varepsilon_4$. Thus the set A is a D -desirable set provided

$$\beta_4 + 4(\omega - \beta_2) + 4(\delta - 3)\varepsilon_4 \geq 1.$$

See Figure 1 for an illustration. $(\square)$

Claim 3 *We may assume every component F in $G_D^\ell[W]$ is either K_2 or C_5 .*

Proof. Assume there is a component F in $G_D^\ell[W]$ that is neither K_2 nor C_5 . Let A be a minimum isolating set of F . By Proposition 1 it holds that $|A| \leq |V(F)|/3$. (Actually, because we may

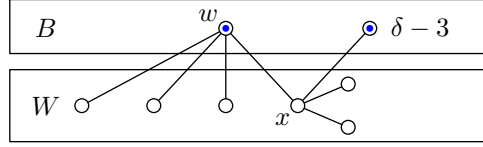


Figure 1: The situation with $\Delta_B = 4$ and $\Delta_W \leq 2$

assume F has maximum degree at most 2, we need this fact only for paths and cycles, and that was shown already by Peters [12].) Since we may assume $\Delta_W \leq 2$ by Claim 2, there are at least $(\delta - 2)|V(F)|$ edges joining F to B . The removal of each such edge decreases the weight of B by at least $\min(\varepsilon_1, \varepsilon_2, \varepsilon_3) = \varepsilon_3$, since by Claim 2 we may assume B_4 is empty. Every vertex in F becomes red. Hence $\xi(A) \geq |V(F)| \times \mathcal{F}_3$ where $\mathcal{F}_3 := \omega + (\delta - 2)\varepsilon_3$. Hence $\xi(A) \geq |V(F)|/3 \geq |A|$ so that A is a D -desirable set provided

$$\omega + (\delta - 2)\varepsilon_3 \geq \frac{1}{3}.$$

See Figure 2 for an illustration. $(\square)$

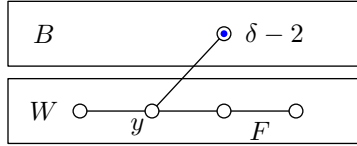


Figure 2: The situation with white component F

Claim 4 *We may assume there is no blue vertex with neighbors in more than one component in $G_D^\ell[W]$.*

Proof. Assume blue vertex x is adjacent to white vertices from two components in $G_D^\ell[W]$, say F_1 and F_2 . By Claim 3, for $i \in \{1, 2\}$ we may assume each F_i is K_2 or C_5 .

Let y_i be a (white) vertex in F_i adjacent to x for $i \in \{1, 2\}$; and for each component F_i that is a 5-cycle, let u_i be a vertex of F_i that is not next to y_i on the cycle. If both F_1 and F_2 are K_2 let $A = \{x\}$; if $F_1 = K_2$ and $F_2 = C_5$ let $A = \{x, u_2\}$; and if both F_1 and F_2 are C_5 let $A = \{x, u_1, u_2\}$. See Figure 3 for an illustration.

In each case A is an isolating set of both the F_i . Thus x and all of $V(F_1) \cup V(F_2)$ become red. Since it is incident with at least two edges, the weight of x decreases by at least β_2 . Hence $\xi(A) \geq |V(F_1) \cup V(F_2)|\omega + \beta_2$. (We note that one can also consider the edges from the F_i to $B \setminus \{x\}$, but this does not help the overall linear program.) Thus A is a D -desirable set provided

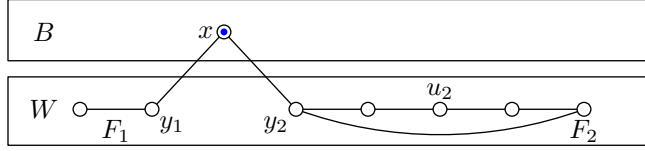


Figure 3: The situation with white component $F_1 = K_2$ and $F_2 = C_5$

$$4\omega + \beta_2 \geq 1,$$

$$7\omega + \beta_2 \geq 2, \text{ and}$$

$$10\omega + \beta_2 \geq 3. \quad (\square)$$

Claim 5 *If D is not an isolating set of G , then there is a D -desirable set.*

Proof. By Claim 3 we may assume that every component F in $G_D^t[W]$ is either K_2 or C_5 . Let B_F be the set of blue vertices adjacent to vertices in F . By Claim 4 we may assume that if $z \in B_F$ then all its neighbors lie in F .

Assume first that $F = K_2$ with vertices v_1 and v_2 , and let $A = \{v_1\}$. The two vertices in F as well as all the vertices in B_F become red. Since F has order 2, we have $B_F \subseteq B_1 \cup B_2$. By the minimum degree, each vertex of F is adjacent to at least $\delta - 1$ blue vertices in G_D^t ; thus there are at least $2(\delta - 1)$ edges joining F to B_F . The removal of an edge joining a vertex in B_1 to F decreases the weight by β_1 , while the collective removal of the two edges joining a vertex in B_2 to F decreases the weight by β_2 . Hence $\xi(A) \geq 2\omega + 2(\delta - 1) \min(\beta_1, \beta_2/2)$. Thus A is a D -desirable set provided

$$2\omega + 2(\delta - 1) \min(\beta_1, \beta_2/2) \geq 1.$$

See Figure 4 for an illustration.

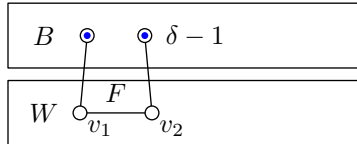


Figure 4: The situation with white component K_2

Assume second that F is the cycle $v_1 v_2 \dots v_5 v_1$, and let $A = \{v_2, v_5\}$. The five vertices in F as well as all the vertices in B_F become red. By Claim 2 we may assume B_4 is empty. By the minimum degree, each vertex of F is adjacent to at least $\delta - 2$ blue vertices in G_D^t ; thus there are at least $5(\delta - 2)$ edges joining F to B_F . The removal of the i edges joining a vertex in B_i to F decreases the weight by β_i in total. Hence $\xi(A) \geq 5\omega + 5(\delta - 2) \min(\beta_1, \beta_2/2, \beta_3/3)$. Thus A is a D -desirable set provided

$$5\omega + 5(\delta - 2) \min(\beta_1, \beta_2/2, \beta_3/3) \geq 2. \quad (\square)$$

This completes the proof of Lemma 1. We provide some consequences.

Theorem 2 *If G is a graph of order n with minimum degree at least 4, then $\iota(G) \leq \frac{13}{41}n$.*

Proof. The system $\mathcal{L}$ is satisfied by setting

$$\omega = \frac{26}{82} \quad \beta_4 = \frac{14}{82} \quad \beta_3 = \frac{12}{82} \quad \beta_2 = \frac{10}{82} \quad \beta_1 = \frac{5}{82}.$$

Thus by Lemma 1 the graph G is w -alluring, and so $\iota(G) \leq \omega n$ by Theorem 1. $\square$

Unfortunately the technique is useless for $\delta = 3$: the optimum for the linear program has $\omega > \frac{1}{3}$ while by Proposition 1 we know that the isolation number is at most $n/3$. But for example for $\delta = 5$, it gives an upper bound of $\frac{23}{78}n$. Note that the $\delta = 4$ and $\delta = 5$ upper bounds are better than the original upper bounds provided in [7].

4 Triangle-Free Graphs and Beyond

Consider the system $\mathcal{L}'$ where the constraints in $\mathcal{L}$ for Claim 5 are replaced by:

$$\begin{aligned} 2\omega + 2(\delta - 1)\beta_1 &\geq 1 & (\text{Claim 6}) \\ 5\omega + 5(\delta - 2) \min(\beta_1, \beta_2/2) &\geq 2 \end{aligned}$$

Lemma 2 *Assume the weights w satisfy the system of constraints $\mathcal{L}'$ for some $\delta \geq 3$. Then every triangle-free graph with minimum degree at least δ is w -alluring.*

The proof of Lemma 2 uses Claims 1 through 4. At that point we may assume that: if F is a component in $G'_D[W]$ then it is either K_2 or C_5 , and any blue vertex adjacent to F is adjacent to no other white component.

Claim 6 *If the weights satisfy the system $\mathcal{L}'$ and D is not an isolating set of G , then there is a D -desirable set.*

Proof. The proof is almost the same as that of Claim 5. The difference is that, by the lack of triangles, we know that when $F = K_2$ each blue vertex has at most one neighbor in F and so $B_F \subseteq B_1$, while when $F = C_5$ each blue vertex has at most two neighbors in F and so $B_F \subseteq B_1 \cup B_2$. Thus the constraints simplify to those given above. $(\square)$

This completes the proof of Lemma 2.

Theorem 3 *If G is a triangle-free graph of order n with minimum degree at least 4, then $\iota(G) \leq \frac{3}{10}n$.*

Proof. The system $\mathcal{L}'$ is satisfied by setting

$$\omega = \frac{3}{10} \quad \beta_4 = \frac{3}{20} \quad \beta_3 = \frac{1}{8} \quad \beta_2 = \frac{1}{10} \quad \beta_1 = \frac{1}{15}.$$

The result follows as before. $\square$

For $\delta \geq 5$ the resultant bound is $\frac{9}{31}n$. Unfortunately this system still does not provide a useful bound for $\delta = 3$, since the solution to the linear program remains bigger than $\frac{1}{3}$. But if we add the requirement that the girth is at least five, then in Claim 5 one gets that for $F = C_5$ each blue vertex has at most one neighbor in F , and so $B_F \subseteq B_1$. If we use this fact to simplify the system of constraints, then

Theorem 4 *If G is a graph of order n with minimum degree at least 3 and girth at least 5, then $\iota(G) \leq \frac{11}{34}n$.*

We note that for 3-regular graphs this upper bound is larger than the upper bound of [13], which holds for all cubic graphs.

5 Lower Bounds

We noted in the introduction that Caro and Hansberg [7] provided graphs of order n with minimum degree δ where the isolation number is $\frac{2}{\delta+4}n$. That family is not regular. For $\delta = 3$ Ziemann and Żyliński [13] found a family of regular graphs with the same ratio $\frac{2}{7}$.

One can use the same approach as [13] to generate some other regular examples. Namely, if F is a regular graph with designated edge xy , then let $G_s(F)$ be the graph formed by taking s copies of the edge-deleted $F - xy$ and adding s edges to make $G_s(F)$ connected and r -regular. See Figure 5 for an illustration.

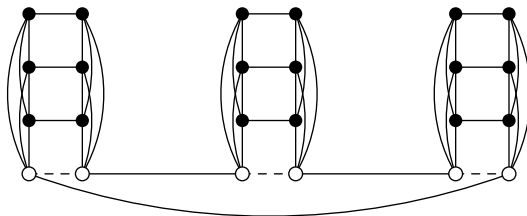


Figure 5: An example graph $G_3(F)$

Lemma 3 *If F has order c and each of F , $F - x$, $F - y$, and $F - \{x, y\}$ have isolation number (at least) b , then the isolation number of $G_s(F)$ is at least $\frac{b}{c}$ its order.*

Proof. Every isolating set of $G_s(F)$ must contain at least b vertices from each copy of F : even if both x and y are dominated from outside, one still needs b vertices to isolate F . Thus $\iota(G_s(F)) \geq bs$ while its order is cs . $\square$

For 4-regular, a suitable F is for example the prism of K_4 with the special edge one joining the two copies of K_4 . This example has $c = 8$ and $b = 2$; so the associated $G_s(F)$ (shown in Figure 5) has isolation number $\frac{1}{4}$ its order. For 4-regular triangle-free graphs, there are several suitable graphs with $c = 14$ and $b = 3$. For 5-regular, there are also many suitable graphs with $c = 14$ and $b = 3$. See Figure 6 for an illustration of two suitable F . In both cases the associated $G_s(F)$ has isolation number $\frac{3}{14}$ its order.

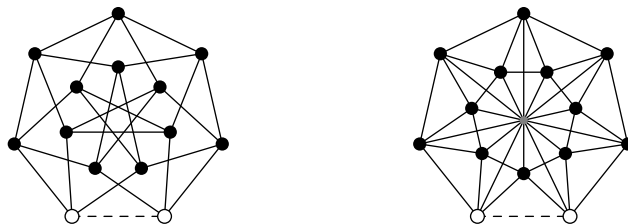


Figure 6: Some suitable F for $G_s(F)$

Acknowledgement

We thank Geoff Boyer for some helpful discussions. Research of Michael A. Henning was supported in part by the South African National Research Foundation, Grant Number 132588, and the University of Johannesburg.

References

- [1] G. Boyer and W. Goddard, Disjoint isolating sets and graphs with maximum isolation number, *Discrete Appl. Math.* 356 (2024), 110–116.
- [2] B. Brešar, S. Klavžar, and D. F. Rall, Domination game played on trees and spanning subgraphs. *Discrete Math.* 313 (2013), 915–923.
- [3] Cs. Bujtás, Domination number of graphs with minimum degree five. *Discuss. Math. Graph Theory* 41 (2021), 763–777.

- [4] Cs. Bujtás and M. A. Henning, On the domination number of graphs with minimum degree six. *Discrete Math.* 344 (2021), Paper 112449.
- [5] Cs. Bujtás and M. A. Henning, On the domination number of graphs with minimum degree at least seven. *Discrete Math.* 348 (2025), Paper 114424.
- [6] Cs. Bujtás and S. Klavžar, Improved upper bounds on the domination number of graphs with minimum degree at least five. *Graphs Combin.* 32 (2016), 511–519.
- [7] Y. Caro and A. Hansberg, Partial domination — the isolation number of a graph. *Filomat* 31 (2017), 3925–3944.
- [8] A. Dapena, M. Lemańska, M.J. Souto-Salorio, and F.J. Vazquez-Araujo, Isolation number versus domination number of trees. *Mathematics* 9 (2021), Paper 1325.
- [9] T. W. Haynes, S. T. Hedetniemi, and M. A. Henning, *Domination in Graphs: Core Concepts* Series: Springer Monographs in Mathematics, Springer, Cham, 2023. xx + 644 pp.
- [10] W. B. Kinnersley, D. B. West and R. Zamani, Extremal problems for game domination number. *SIAM J. Discrete Math.* 27 (2013), 2090–2107.
- [11] B. Krishnakumari, Y.B. Venkatakrishnan, and M. Krzywkowski, Bounds on the vertex-edge domination number of a tree. *C. R. Math. Acad. Sci. Paris*, 352 (2014), 363–366.
- [12] J.W. Peters, Theoretical and Algorithmic Results on Domination and Connectivity. Ph.D. Thesis, Clemson University, Clemson (1986).
- [13] R. Ziemann and P. Żyliński, Vertex-edge domination in cubic graphs. *Discrete Math.* 343 (2020), Paper 112075.
- [14] P. Żyliński, Vertex-edge domination in graphs. *Aequationes Math.* 93 (2019), 735–742.